

Q1: The following C++ program is designed to find the missing number from an array (unordered) containing integers from 0 to $n-1$, where exactly one number is missing. [Marks 10]

Task: Complete the missing code segments so that the program correctly identifies and prints the missing number. Write your answers only in the provided spaces. Enough space is provided for answers. Note: Assume iostream header file and main function is provided.

Restrictions:

1. Do not create additional arrays or variables (both).
2. Do not use nested loops.
3. Code must be generic (work for any n).
4. Bitwise operators not allowed.

```
const int n = 9;  
int arr[n - 1] = {1, 0, 5, 2, 4, 8, 7, 6}; // missing number is 3
```

```
int mystery = 0;
```

```
for (int i = 0; i < n; i++) {
```

```
<< endl;
```

```
cout << "Missing number is: " << _____
```

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[CLO1: Demonstrate the basic concepts of programming]

Q2: The following C++ program is designed to remove all occurrences of a given substring (char array) from a string (char array). [Marks 10]

Task: Complete the missing code segments so that the program correctly removes every occurrence of the substring and displays the final string. Write your answers only in the provided spaces. Enough space is provided for answers.

Restrictions:

1. Code must be generic (independent of array sizes).
2. Do not create additional arrays or variables (both).
3. Do not use other header files (string, algorithm, cstring etc.). Built-in functions are not allowed.

```
#include <iostream>
using namespace std;
int main() {
    char str[50], sub[10];
    cout << "Enter main string: ";
    cin >> str; // input: abcXabcabYcZabcAabc
    cout << "Enter substring to remove: ";
    cin >> sub; // input: abc
    int p = 0, q = 0;

    while ( p < 10 ) {
        int i = 0;
        while ( q < 50 ) {
            if ( str[i] == sub[p] )
            if ( sub[p] == str[q] ) {
                i = q;
                for ( i i < 50; i++ ) {
                    str[i] = str[i+1]; } }
            q++;
            q++;
            p++;
        }
    }

    str[q] = '\0';
    cout << "String after removal: " << str << endl; // output: XabYcZA
    return 0;
}
```

LO1: Demonstrate the basic concepts of programming]

Q3: The following incomplete C++ program performs different bitwise operations on a 32-bit unsigned integer inspired from Assignment 2 question Holocron of Malachor. It uses only bitwise operators and switch-case statements (no loops, no if-else, no logical operators). Your task is to complete the code in switch cases as mentioned in the comments as per given input and output. [Marks 10]

```
#include <bitset>
#include <iostream>
using namespace std;
int main() {
    unsigned int holocron, result;
    int choice;
    cin >> choice;
    switch (choice) {
        case 1: // Reverse all bits (32-bit version) [3 Marks]
            result = holocron; // result = 196758
            cout<<result<<'\t'<<bitset<32>(result)<<endl;
            //196758          0000000000000000110000000010010110
```

```
for (int i=2; i<32; i++) {
    int bit = (result) & 1;
    result = result >> i;
    result = result | (bit << 31);
}
```

```
cout<< result<<'\t'<<bitset<32>( result)<<endl;
//1761656832      01101001000000001100000000000000
```

break;

```
case 2: // Circular left shift by 1 bit (32-bit version) [2 Marks]
    result = holocron; // result = 2147483648
    cout<<result<<'\t'<<bitset<32>( result)<<endl;
    //2147483648      10000000000000000000000000000000
```

```
result = (result << 1) | ((result & 1) << 31);
```

```
cout<<result<<'\t'<<bitset<32>( result)<<endl;
// 1          00000000000000000000000000000001
```

break;

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case 3: // Count number of 1-bits (parallel popcount, 32-bit version)

[3 Marks]

```
result = holocron; // result = 196758
cout<<result<<'\\t'<<bitset<32>( result)<<endl;
//196758      000000000000000110000000010010110
int count = 0
```

```
for (int i=0; i<32; i++) {
```

```
}
```

```
cout<<"No. of 1 bits= "<<result<<endl;
//No. of 1 bits= 6
```

```
break;
```

case 4: // Check MSB (32 bit version) [1 Mark]

```
result = holocron; // result = 2147483648
cout<<result<<'\\t'<<bitset<32>( result)<<endl;
//2147483648  10000000000000000000000000000000
```

```
switch (static_cast<bool>(result & (1 << 31))) {
    case 0: cout << "Seal Intact"; break;
    case 1: cout << "Seal Broken"; break;
} // Seal Broken
break;
```

case 5: // Bitwise NOT (32 bit version) [1 Mark]

```
result=holocron; // result = 196758
cout<<result<<'\\t'<<bitset<32>( result)<<endl;
//196758      000000000000000110000000010010110
```

```
result = ~result;
```

```
cout<<result<<'\\t'<<bitset<32>( result)<<endl;
//4294770537  11111111111111001111111101101001
```

```
break;
```

```
}
return 0;
```

CLO1: Demonstrate the basic concepts of programming]

Q4: For each code segment below, write only the final output in the space provided.

- Do not write anything other than the output.
- No cutting, overwriting, or corrections are allowed.
- Assume all programs run without any errors i.e. Assume iostream header file and main function is provided.

[Total: 30 marks — marks for each part are mentioned on the right]

	Code	Output	Marks
1	<pre>int i = 2, j = 3, k = 4, l = 5; cout << j + (++l) -- i++ + k << endl;</pre> <i>3 + 6 - 2 + 4</i>	30	1
2	<pre>int i = 2, j = 3, k = 4, l = 5; cout << i - j++ + (--k) -- * ++l << endl;</pre> <i>2 - 3 + 3 * 6</i>	17	1
3	<pre>int i = 2, j = 3, k = 4, l = 5; cout << --j - ++i + l + k -- << endl;</pre> <i>2 - 4 + 3 + 4</i>	7	1
4	<pre>int i = 2, j = 3, k = 4, l = 5; cout << (((++i)++ + --l) - (++j + k--)) << endl;</pre> <i>(3 + 4) - (4 + 5)</i>	9	1
5	<pre>int testarray[3][2][2] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}; cout << testarray[2][1][0] << endl;</pre>	11	2
6	<pre>int testarray[][3][2] = {{1, 2}, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13}; cout << testarray[2][2][1] << endl;</pre>	0	2
6	<pre>int i = 9, counter = 0; while ((i-1)) { ++counter; i--; } cout << counter;</pre> <i>1 2 3 4 5 6 7 8</i> <i>8 7 6 5 4 3 2 1</i>	8	2
7	<pre>int i, j; int counter = 0; int arr[8][9]; for (i=0; i<9; i++) for (j=0; j<8; j++) { arr[j][i] = counter; ++counter; } cout << arr[6][8];</pre>	70	2
8	<pre>int n=0; int mystery=4; while(n++ / mystery < 3) { if(n % mystery == 0) n = n/mystery; cout << "Hello !! " << endl; n++; }</pre>	?	3

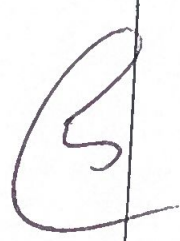
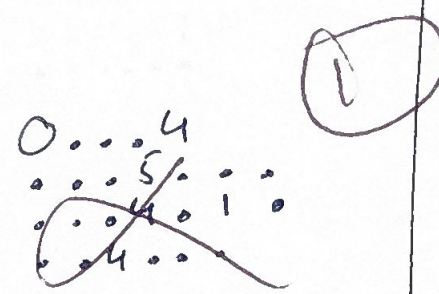
2 2
3 4

2 6
7 8

2 10
11 12

2 2
0 0
0 0

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9 <pre> int n = 0; while (++n / 5 < 4) { if (n % 7 == 0) { ++n += 3; continue; } if (n / 3 == 5) { break; } cout << "Hello !! " << n << endl; n++; } </pre>	Hello!! 1 Hello!! 3 Hello!! 5 Hello!! 12 Hello!! 19 	5
6 <pre> int z = 5; int a = z / 2; for(int t = 0; t < z * z; t++) { int p = t / z; int q = t % z; int s = p - a; if (s < 0) s = -s; int easy = a - s; int hard = a + s; if (q == easy q == hard) cout << ((p + q) % 7); else cout << "."; if (q == z - 1) cout << endl; } </pre>		5
7 <pre> int n = 5; char arr[6][6] = {}; for (int i = 1; i <= n; i++) { for (int j = 1; j <= n; j++) { arr[i][j] = (i == j i + j == n + 1) ? '*' : (i == 1 i == n j == 1 j == n) ? '-' : ((i == n / 2 + 1 && j != 1 && j != n) (j == n / 2 + 1 && i != 1 && i != n)) ? '+' : '.'; } } for (int i = 1; i <= n; i++) { for (int j = 1; j <= n; j++) { cout << arr[i][j]; } cout << endl; } </pre>		5